

CHAPTER 6.1: INTRODUCTION TO PRIVACY

FOUNDATIONS OF DISTRIBUTED SYSTEMS

Isabel Wagner

Fall semester 2025

University of Basel

1. Privacy violations: examples
2. Privacy definitions and taxonomies
3. Privacy engineering
4. Data protection regulation
5. Who owns privacy?

1. PRIVACY VIOLATIONS: EXAMPLES

PRIVACY VIOLATIONS: EXAMPLE I (CAMBRIDGE ANALYTICA)

- In 2014, Cambridge Analytica published a personality test *“This Is Your Digital Life”* on Facebook
 - Objective: collect data for (allegedly) scientific purposes
 - The app had ~320'000 users
 - Each user agreed to release all their personal data from Facebook to Cambridge Analytica (informed consent – even though hidden in the terms and conditions)
- But: Facebook’s design allowed this app not only to collect the personal information of people who agreed to take the survey, but also the personal information of all the people in those users’ Facebook social network
 - Data from ~50 million users (some sources even report ~87 million)
 - Data has been used for micro-targeting campaigns in politics (e.g., for the Brexit vote in 2016 and D. Trump’s presidential campaign)

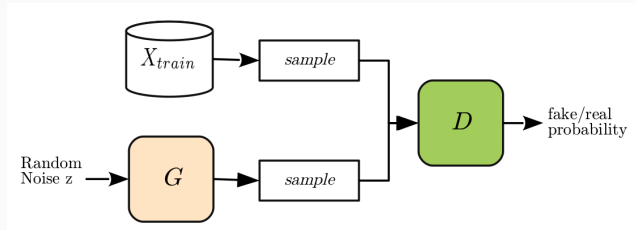
PRIVACY VIOLATIONS: EXAMPLE II (NETFLIX)

- In 2006, Netflix started a \$1 million contest crowdsourcing campaign to improve their recommendation system
 - Basis: **users' view histories and user ratings**, released by Netflix
 - Netflix provided training data set of 100'480'507 ratings from 480'189 users on 17'770 movies
 - Anonymized training data: user, movie, date of grade, grade, where user and movie fields are integers
- Based on film ratings on IMDB, **quite a number of individual users could be identified (de-anonymized)**¹
- ...and thus their complete view history revealed
- Surprisingly, the de-anonymization attack succeeded even if the data was **k-anonymized** before running the attack

¹A. Narayanan and V. Shmatikov, "Robust de-anonymization of large sparse datasets," in *IEEE Symposium on Security and Privacy, 2008*, Oakland, CA, USA: IEEE, May 2008, pp. 111-125. doi: 10.1109/SP.2008.33. [Online]. Available: http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=4531148.

PRIVACY VIOLATIONS: EXAMPLE III (MACHINE LEARNING)...

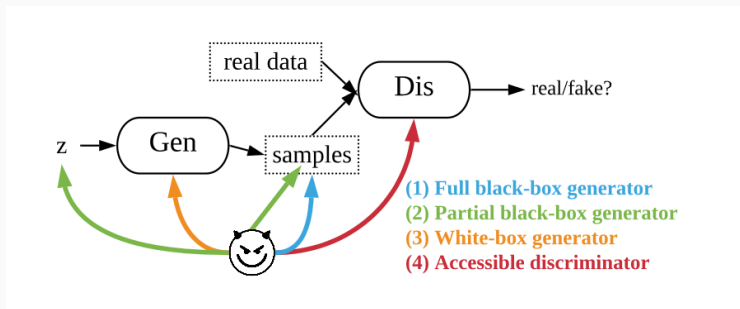
- If we can't safely release datasets, how about we release synthetic data?
- Use data to train a model and make predictions based on this model²
 - Generator model G generates samples based on random noise, learns to generate better samples based on fake/real probability from D
 - Discriminator model D learns to tell which samples are from training data, which are generated
 - Released data = samples from G . So all is good, privacy-wise?



²J. Hayes, L. Melis, G. Danezis, et al., "LOGAN: Membership Inference Attacks Against Generative Models," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 1, pp. 133–152, Jan. 2019. DOI: 10.2478/popets-2019-0008.

...PRIVACY VIOLATIONS: EXAMPLE III (MACHINE LEARNING)

- Unfortunately not: Privacy of individuals in the training data is still at risk.
- **Membership inference**: adversary can tell if a specific known data point (or a specific known data set) was contained in the training data
- Many different attacks proposed, depending on what the adversary can do³:



³D. Chen, N. Yu, Y. Zhang, et al., "GAN-Leaks: A Taxonomy of Membership Inference Attacks against Generative Models," in *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, ser. CCS '20, New York, NY, USA: Association for Computing Machinery, Oct. 2020, pp. 343–362. DOI: 10.1145/3372297.3417238.

PRIVACY VIOLATIONS: EXAMPLE IV (ZOOM)

- A motherboard report found Zoom sends data from users of its iOS app to Facebook for advertising purposes⁴
- From the Zoom Privacy Policy (February 2020, has been updated since)
 - “Does Zoom sell Personal Data?
Depends what you mean by ‘sell’. We do not allow marketing companies, advertisers, or anyone else to access Personal Data in exchange for payment. So in our humble opinion, we don’t think most of our users would see us as selling their information [...] Sharing Personal Data with the third-party provider while using these tools may fall within the extremely broad definition of the ‘sale’ of Personal Data under certain state laws because those companies might use Personal Data for their own business purposes [...].”
 - “We may transfer your Personal Data to the U.S., to any Zoom affiliate worldwide, or to third parties acting on our behalf for the purposes of processing or storage.”

⁴<https://www.theguardian.com/technology/2020/apr/08/zoom-privacy-video-chat-alternatives>

PRIVACY VIOLATIONS: EXAMPLE V (THE USE OF OUTSIDE INFORMATION)

- Early systematic linking of data sources in the **1970s** in Germany:
 - “**Rasterfahndung**” (dragnet investigation) introduced by Horst Herold, Head of the Bundeskriminalamt (German FBI equivalent)
 - Approach: intersection of datasets to extract short list of persons to observe.
 - For example: *recent renter $\cap$ pays electricity bill in cash $\cap$ less than 15km from kidnapping location $\cap$ close to a highway exit*
 - Successful in several cases, but failed in one crucial case (Schleyer) because one observation record of suspicious activity was lost.
 - Article from Michael Jürgs: “Der letzte Gefangene der RAF – Horst Herold wird 95”, Oct 21, 2018⁵
- Does this violate the privacy of terrorists?
- Does it violate the privacy of innocent citizens who happen to meet the criteria?. At stake here: **collection of metadata**.

⁵<https://www.tagesspiegel.de/gesellschaft/panorama/der-letzte-gefangene-der-raf-3999820.html>

PRIVACY VIOLATIONS: EXAMPLE V (THE USE OF OUTSIDE INFORMATION, CONTD)

- Note: Encryption provides confidentiality, but not privacy.
- Former head of CIA and NSA, General Hayden, in 2014: “We kill people based on Metadata”
- Many research papers on ML classification (“fingerprinting”) of encrypted VoIP calls, encrypted web traffic
- Metadata collection is big business!
 - NY Times article on location data collection by mobile apps⁶
 - MapQuest’s “Location Intelligence”⁷:

MapQuest first party consumer location data consists of over 30 million unique visitors’ keyword search, route search, route taken, POI visitation and mobile GPS data per month. Leverage this movement data to make smarter, more strategic business decisions about your offerings and operations.

MapQuest users must approve the use of their data by opting into the program, and all users’ data is anonymized to ensure the privacy of our loyal users.

⁶<https://www.nytimes.com/interactive/2018/12/10/business/location-data-privacy-apps.html>

PRIVACY VIOLATIONS: EXAMPLE VI (FACIAL RECOGNITION)

- Facial processing technology (face detection + verification + identification + facial analysis) erodes privacy⁸
 - Face data is **biometric** ⇒ unique, identifiable information
 - Face data casually available in many forms ⇒ perpetuates **severe privacy violations**:
 - Data sometimes collected **without consent** in online downloadable face datasets
 - Female subjects included in datasets with consent but **not displayed/distributed publicly** due to cultural restrictions of their exposure (e.g., Iranian dataset)
 - Hunt for increased demographic diversity may violate privacy: Images of “baby”, “kid”, “teenager” on Google image search to supplement datasets with mainly adult subjects, **rarely asking for parental consent**
 - DeepFace (DL model, dataset not released) replaces manual verification and labeling ⇒ auto-generated labels with **offensive terminology**

⁸<https://www.technologyreview.com/2021/02/05/1017388/ai-deep-learning-facial-recognition-data-history/>

PRIVACY VIOLATIONS: EXAMPLE VI (FACIAL RECOGNITION)

Facial recognition data source distribution by era

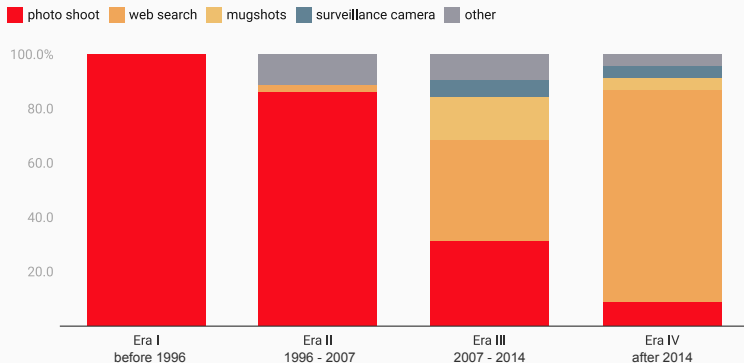


Chart: MIT Technology Review • Source: [Raji & Fried](#) • Created with [Datawrapper](#)

“Dataset creation through image retrieval at scale on the web raises serious concerns around privacy, ownership, and consent—a legal question that [computer scientists](#) should also be actively engaging.”⁹

⁹I. D. Raji and G. Fried, *About Face: A Survey of Facial Recognition Evaluation*, Feb. 2021. DOI: 10.48550/arXiv.2102.00813. arXiv: 2102.00813 [cs]

2. PRIVACY DEFINITIONS AND TAXONOMIES

*“No one shall be subjected to arbitrary interference with his **privacy**, family, home or correspondence, nor to attacks upon his honour and reputation.*

*Everyone has the **right** to the protection of the law against such interference or attacks”*

(Article 12, Universal Declaration of Human Rights (1948))

WHY IS PRIVACY A HUMAN RIGHT?

- Consider privacy in the context of other human rights: equality (art. 2), no torture (art. 5), freedom of opinion (art. 19)
- Consider the common response: “I have nothing to hide!”

“Saying you don’t care about privacy because you have nothing to hide is like saying you don’t care about free speech because you have nothing to say”
(Edward Snowden, 2019)

- Without privacy...
 - We can’t enjoy other rights, e.g., freedom of thought, freedom of assembly¹⁰¹¹
 - We can’t develop a sense of identity
 - We have limited human autonomy and self-determination

¹⁰G. Buttarelli, “Privacy matters: Updating human rights for the digital society,” *Health and Technology*, vol. 7, no. 4, pp. 325–328, Dec. 2017. DOI: 10.1007/s12553-017-0198-y. [Online]. Available: <https://doi.org/10.1007/s12553-017-0198-y>.

¹¹S. Wachter, “Privacy: Primus Inter Pares - Privacy as a Precondition for Self-Development, Personal Fulfilment and the Free Enjoyment of Fundamental Human Rights,” Social Science Research Network, Rochester, NY, SSRN Scholarly Paper ID 2903514, Jan. 2017. [Online]. Available:

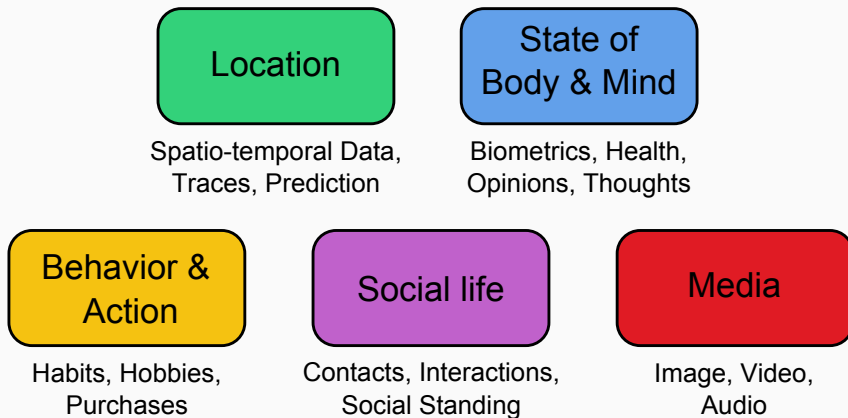
WHAT DOES IT MEAN TO HAVE PRIVACY?

Many definitions proposed in the last 130 years:

- Right to be let alone (Warren and Brandeis, 1890)
- Right to control information about oneself (Westin, 1970)
- The opposite of privacy: activities that are harmful to privacy (Solove, 2006)
- Privacy as contextual integrity (Nissenbaum, 2010)
 - Privacy means that information flows follow contextual information norms
 - Contextual information norms: standards regarding information flow that society wants to uphold in specific contexts
 - For example: medical data shared with a doctor remain confidential, i.e., the data do not flow to the patient's employer
- Privacy is a “concept in disarray” and “nobody can articulate what it means” (Solove, 2006¹²)

¹²D. J. Solove, “A Taxonomy of Privacy,” *University of Pennsylvania Law Review*, vol. 154, no. 3, pp. 477–564, Jan. 2006. DOI: 10.2307/40041279.

- It is not 2006 anymore.
- As computer scientists, we need privacy definitions and taxonomies that help us design, build, analyze computer systems that protect privacy
- We can consider different aspects separately:
 1. Privacy types (data types, contexts, scenarios)
 2. Adversaries (privacy against who)
 3. Privacy properties and privacy threats
 4. Activities that can harm privacy



¹³D. Eckhoff and I. Wagner, "Privacy in the Smart City – Applications, Technologies, Challenges and Solutions," *IEEE Communications Surveys & Tutorials*, vol. 20, no. 1, pp. 489–516, 2018. DOI: 10.1109/COMST.2017.2748998.

Adversary motivations

- Fun, curiosity
- Profit
- Crime (also detection, prevention)
- Semi-honest: follows the protocol, but wants to learn additional information
- Malicious: does not necessarily follow the protocol

{ human-level motivations

{ abstractions for analysis

Adversary capabilities

- Active – passive
- Static – adaptive
- Access to resources: computational power, network bandwidth
- Access to algorithms: maybe computationally bounded, e.g., restricted to complexity class PP: solvable by probabilistic Turing machine in polynomial time

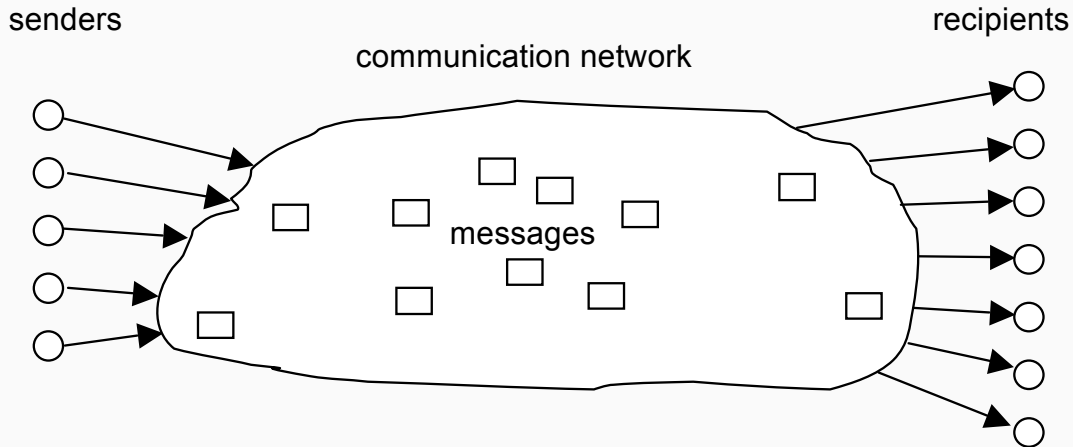
Adversary locations

- Global – local
- Internal – external
- Service provider – device manufacturer – software developer – third party

	Privacy properties	Privacy threats
HARD	Unlinkability	Linkability
	Anonymity & Pseudonymity	Identifiability
	Plausible deniability	Non-repudiation
	Undetectability & Unobservability	Detectability
	Confidentiality	Disclosure of information
SOFT	Content awareness	content Unawareness
	Policy and consent compliance	policy and consent Noncompliance

LINDDUN: framework for analyzing privacy threats

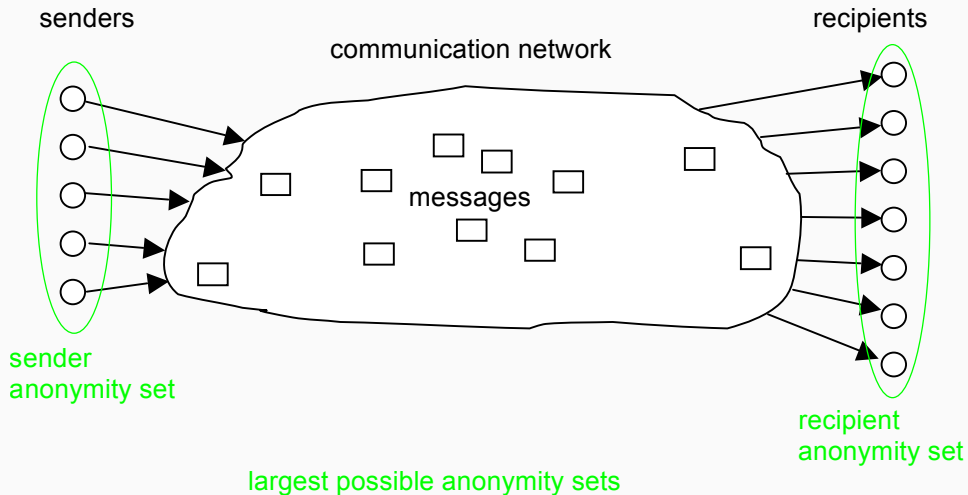
¹⁴M. Deng, K. Wuyts, R. Scandariato, *et al.*, “A privacy threat analysis framework: Supporting the elicitation and fulfillment of privacy requirements,” *Requirements Engineering*, vol. 16, no. 1, pp. 3–32, Mar. 2011. DOI: 10.1007/s00766-010-0115-7. [Online]. Available: <https://link.springer.com/article/10.1007/s00766-010-0115-7>.



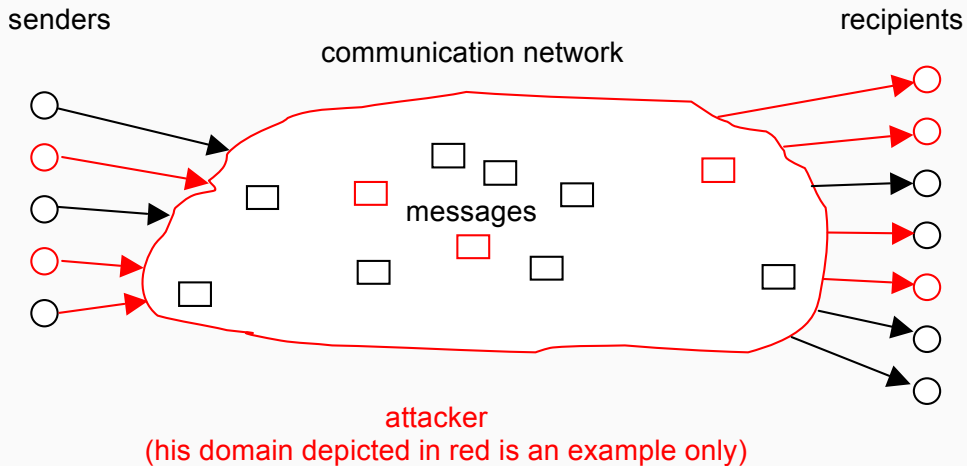
¹⁵A. Pfitzmann and M. Hansen, "A terminology for talking about privacy by data minimization: Anonymity, Unlinkability, Undetectability, Unobservability, Pseudonymity, and Identity Management," Tech. Rep., Aug. 2010. [Online]. Available: http://dud.inf.tu-dresden.de/literatur/Anon_Terminology_v0.34.pdf.

ANONYMITY SETS

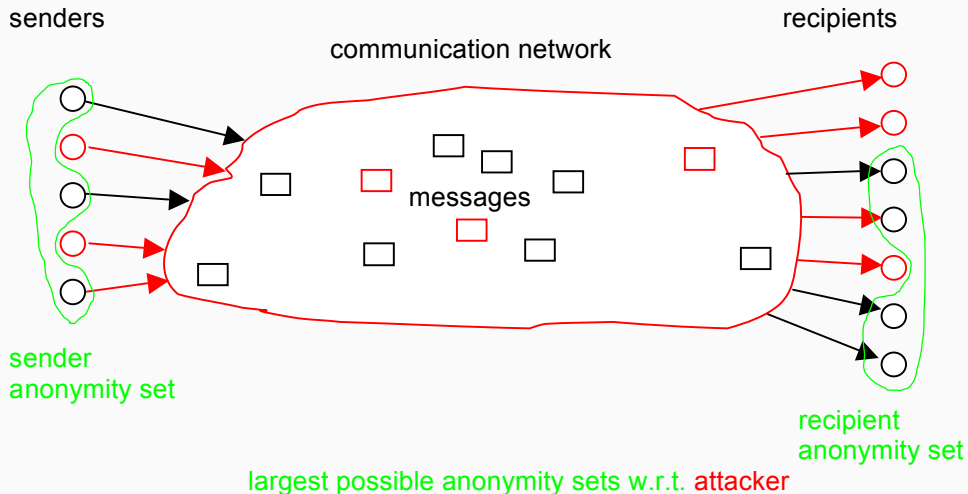
Anonymity: subject is not identifiable within a set of subjects



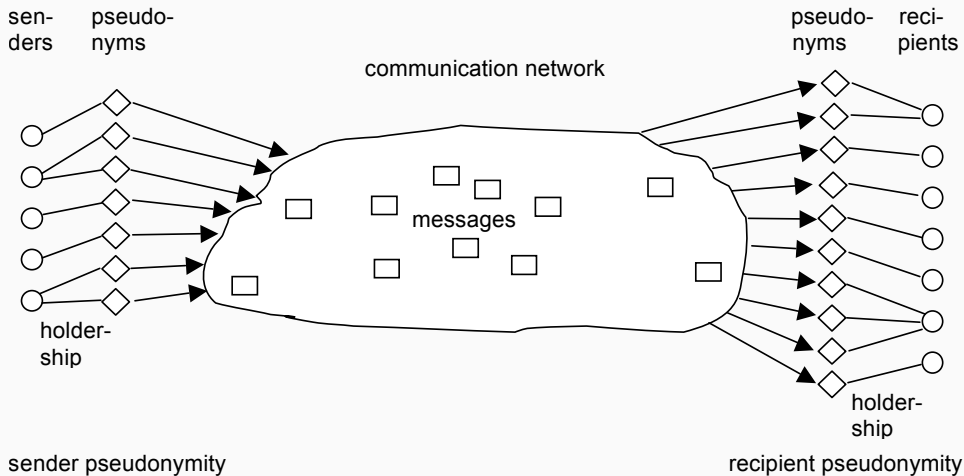
ANONYMITY, CONSIDERING THE ATTACKER



ANONYMITY SETS, CONSIDERING THE ATTACKER



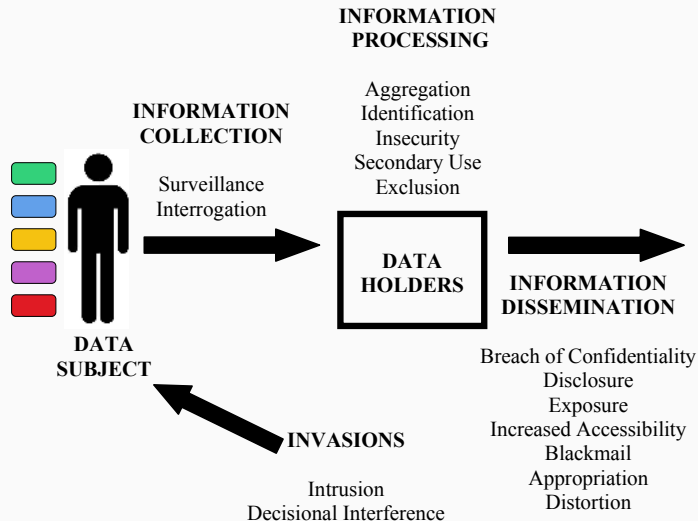
PRIVACY PROPERTY: PSEUDONYMITY¹⁶



¹⁶A. Pfitzmann and M. Hansen, "A terminology for talking about privacy by data minimization: Anonymity, Unlinkability, Undetectability, Unobservability, Pseudonymity, and Identity Management," Tech. Rep., Aug. 2010. [Online]. Available: http://dud.inf.tu-dresden.de/literatur/Anon_Terminology_v0.34.pdf.

- **Unlinkability**: attacker cannot distinguish whether two items of interest (e.g., subjects, messages) are related
- **Undetectability**: attacker cannot distinguish whether an item of interest exists
- **Unobservability**: undetectability of an item of interest + anonymity of all involved subjects

ACTIVITIES THAT CAN HARM PRIVACY¹⁷



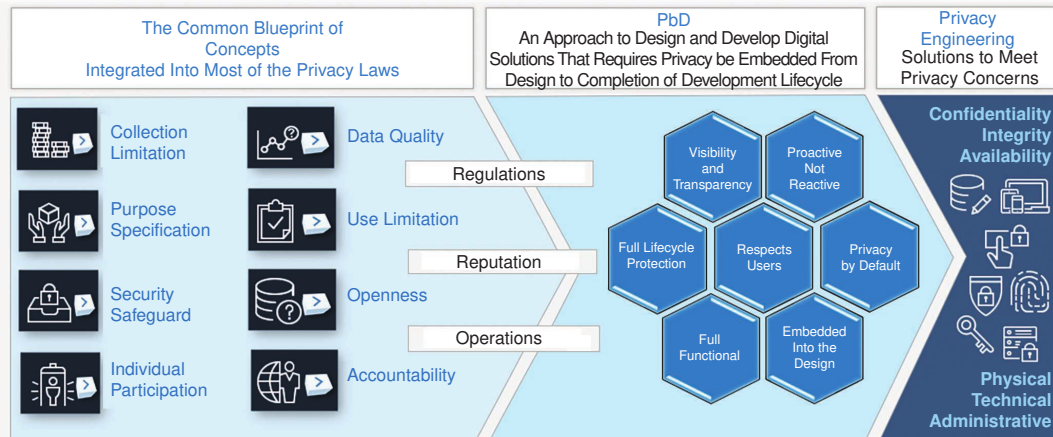
¹⁷D. J. Solove, "A Taxonomy of Privacy," *University of Pennsylvania Law Review*, vol. 154, no. 3, pp. 477–564, Jan. 2006. DOI: 10.2307/40041279.

3. PRIVACY ENGINEERING

- Now that we have **defined** privacy – how do we make sure that computing systems **provide** privacy?
- Develop the tools, methodologies, and processes for meeting the privacy requirements and expectations of regulators and customers
- Create technical solutions based on techniques for data protection and cybersecurity

Fair information principles

Privacy by Design



¹⁸J. Williams and L. Nee, "Privacy Engineering," *Computer*, vol. 55, no. 10, pp. 113–118, Oct. 2022. DOI: 10.1109/MC.2022.3192948.

FAIR INFORMATION PRINCIPLES (FIPS)

- 8 principles, posed by the Organisation for Economic Co-operation and Development (OECD) in 1980¹⁹
- FIPs are intended to be integrated into national privacy law
 - Part of the EU's GDPR
 - and Swiss new Act on Federal Data Protection (nFADP)
- Note: FIPs are **different** from the fair information practice principles (FIPPs) posed by the Federal Trade Commission in the US!²⁰

¹⁹OECD, “Recommendation of the Council concerning Guidelines Governing the Protection of Privacy and Transborder Flows of Personal Data,” Tech. Rep. OECD/LEGAL/0188, 1980.

²⁰Federal Trade Commission, *Privacy Online: A Report to Congress*, <https://www.ftc.gov/reports/privacy-online-report-congress>, 1998.

1. **Collection Limitation.** Data collectors should only collect information that is **necessary**, and should do so by **lawful** and **fair** means, i.e., with the knowledge or **consent** of the data subject.
2. **Data Quality.** The collected data should be kept **up-to-date** and stored only as long as it is **relevant**.
3. **Purpose Specification.** The purpose for which data is collected should be **specified** (and announced) **ahead** of the data collection.
4. **Use Limitation.** Personal data should only be used for the **stated purpose**, except with the data subject's consent or as required by law.

²¹OECD, "Recommendation of the Council concerning Guidelines Governing the Protection of Privacy and Transborder Flows of Personal Data," Tech. Rep. OECD/LEGAL/0188, 1980.

5. **Security Safeguards.** Reasonable security safeguards should protect collected data from unauthorized access, use, modification or disclosure.
6. **Openness.** It should be possible for data subjects to learn about the data controller's identity, and how to get in touch with them.
7. **Individual Participation.** Data subjects should be able to query data controllers whether or not their personal information has been stored, and, if possible, challenge (i.e., erase, rectify, or amend) these data.
8. **Accountability.** Data controllers should be accountable for complying with these principles.

²²OECD, "Recommendation of the Council concerning Guidelines Governing the Protection of Privacy and Transborder Flows of Personal Data," Tech. Rep. OECD/LEGAL/0188, 1980.

7 Foundational Principles:

1. Proactive not reactive; preventive not remedial
2. Privacy as the default setting
3. Privacy embedded into design
4. Full functionality – positive-sum, not zero-sum
5. End-to-end security – full lifecycle protection
6. Visibility and transparency – keep it open
7. Respect for user privacy – keep it user-centric

²³A. Cavoukian, “**Privacy by Design**,” Information and Privacy Commissioner of Ontario, Tech. Rep., 2013. [Online]. Available: <https://www.ipc.on.ca/wp-content/uploads/2013/09/pbd-primer.pdf>.

4. DATA PROTECTION REGULATION

DATA PROTECTION LEGISLATION WORLDWIDE²⁴

71%

COUNTRIES WITH
LEGISLATION

9%

COUNTRIES WITH
DRAFT LEGISLATION

15%

COUNTRIES WITH
NO LEGISLATION

5%

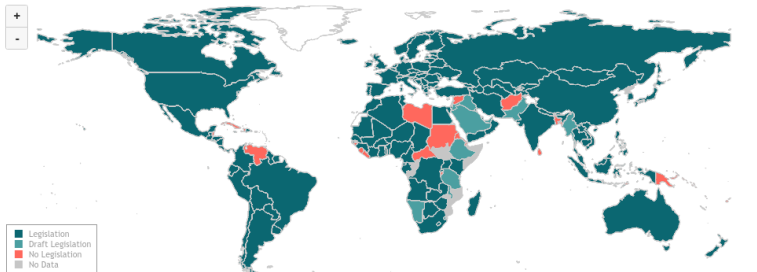
COUNTRIES WITH
NO DATA

NOTHING SELECTED

EUROPE (45 COUNTRIES)

DOWNLOAD FULL DATA

Data Protection and Privacy Legislation Worldwide



Source: UNCTAD, 14/12/2021

Europe (45 countries)

Countries with legislation

Legislation:
44 (98%)

Draft Legislation:
0 (0%)

No Legislation:
0 (0%)

No Data:
1 (2%)

²⁴UNCTAD, *Data Protection and Privacy Legislation Worldwide*, <https://unctad.org/page/data-protection-and-privacy-legislation-worldwide>, 2021.

- **Motivation:** protect fundamental rights and freedoms of natural persons, simplify free movement of personal data within the Union
 - “The protection of natural persons in relation to the processing of personal data is a fundamental right.”
 - “The right to the protection of personal data is not an absolute right; it must be considered in relation to its function in society and be balanced against other fundamental rights, in accordance with the principle of proportionality.”
- **Scope:** applies to processing of personal data when data controller or data processor are in the EU, or when data subject is in the EU or an EU citizen

²⁵European Parliament and Council of the European Union, *General Data Protection Regulation*, <https://eur-lex.europa.eu/eli/reg/2016/679/oj>, Apr. 2016.

- **Personal data:** any information relating to an **identified or identifiable** natural person
 - Direct identifiers: name, identification number, location data, online identifier
 - Indirect identifiers: one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity
- **Data processing:** collection, recording, organization, structuring, storage, adaptation or alteration, retrieval, consultation, use, disclosure by transmission, dissemination or otherwise making available, alignment or combination, restriction, erasure or destruction
- **Controller:** natural or legal person that determines why and how data is processed
- **Processor:** natural or legal person that performs the processing (→ outsourcing)

LAWFUL DATA PROCESSING UNDER THE GDPR (ART. 5–11)

- Principles of data processing: follows the FIPs
- Personal data must be processed **lawfully**, i.e., only if one of these legal bases are met:
 - Data subject has given consent (freely given, specific, informed and unambiguous + withdrawing consent must be as easy as giving it!)
 - To fulfill contractual obligations with a data subject
 - To comply with data controller's legal obligations
 - To protect the vital interests of the data subject or other natural persons
 - To perform a task in the public interest
 - For the legitimate interests of a data controller or a third party, unless these interests are overridden by interests of the data subject or her or his rights, especially children
- Additional rules for **special category** data, e.g., racial or ethnic origin, political opinions, religious or philosophical beliefs, genetic data, biometric data, data concerning health

RIGHTS OF DATA SUBJECTS (ART. 12–22)

- **Transparency:** communication from the data controller must be concise, transparent, intelligible, easily accessible, using clear and plain language
- **Right of access:** data subjects can ask for copy of their data and information on how it is processed
- **Right to rectification:** controllers must correct incorrect data
- **Right to erasure:** data subjects can ask for data deletion under specific circumstances, e.g., withdrawn consent, unlawful processing
- **Right to data portability:** data must be provided in machine-readable form, also to transfer to another controller *where technically feasible*
- **Right to object:** data subject can object to data processing, in particular if purpose of processing is marketing or profiling
- **Right not to be subject to automated decision-making**

OBLIGATIONS FOR CONTROLLERS AND PROCESSORS (ART. 23–43)

- **Data protection by design and by default:** following the state of the art and principles like data minimization; also state-of-the-art security!
- Inform data subjects of their rights, the specifics of data collection and processing, and whether data is transferred to third parties in/outside of EU
- Notify supervisory authority of data breaches within 72 hours, notify data subject **without undue delay**
- Carry out a **data protection impact assessment** for high-risk processing, in particular when using new technologies
- Have a **data protection officer** to monitor and advise on compliance with GDPR

WHAT HAPPENS IN CASE OF GDPR VIOLATIONS? (ART. 77–84)

- Data subjects can complain to the supervisory authority
- Organizations (e.g., NGOs) can represent data subjects in complaints
- Supervisory authority can issue fines for GDPR violations: up to €20 million or up to 4% of the annual worldwide turnover (whichever is greater)



Meta, 2023 – **€1.2 billion**: transferring data from EU Facebook users to the US (violating Schrems II decision)



Amazon, 2021 – **€746 million**: tracking user data without appropriate consent from users or providing the means to opt out



Meta (Instagram) – **€405 million**: wrongfully processing the personal data of children



Meta (Facebook and Instagram), 2023 – **€390 million**: reliance on “contract” as their legal basis in a way that was unclear to users



TikTok, 2023 – **€345 million**: wrongfully collecting the personal data of children and setting their accounts to public by default



Meta (Facebook), 2022 – **€265 million**: making a dataset of personal information available to the internet

²⁶<https://termly.io/resources/articles/biggest-gdpr-fines/>

5. WHO OWNS PRIVACY?

BENTHAM'S PANOPTICON

An architectural solution to a societal problem, the “ideal prison”

- Proposed in 1789
- Analyzed in 1975 by Michel Foucault “Surveiller et punir” (Discipline and Punish - The Birth of the Prison), has no reference to computers
- 20 years later: Link to computers and Internet in 1994, David Lyon “From Big Brother to Electronic Panopticon”
- 2014-2019 (Shoshana Zuboff) : “The age of surveillance capitalism”



BENTHAM'S PANOPTICON

Panopticon prison buildings still exist, but often abandoned:



Stateville, Illinois, still in use. Cells have bars instead of doors²⁷



Presidio modelo, Cuba, abandoned 1967. Built for 500 prisoners per building²⁸

²⁷<https://prisonphotographystaging.wordpress.com/tag/david-leventi/>

²⁸<https://www.thevintagenews.com/2016/06/27/cubas-abandoned-panopticon-prison-prisoners-watched-minimum-staff/>

PRISON GUARDS – WHY MENTION THE PANOPTICON?

- **Lack of privacy seen as a positive property**
 - Bentham: *“Morals reformed - health preserved - industry invigorated, instruction diffused - public burthens lightened - Economy seated, as it were, upon a rock - the gordian knot of the Poor-Laws are not cut, but untied - all by a simple idea in Architecture!”*
- Today, this idea is not limited to prisons:
 - Privacy as friction for the economy, think Silicon Valley
 - Privacy as friction for law enforcement and agencies, think NSA/Snowden
 - Privacy as friction for the environment, think temp/humidity sensors in Swiss apartments
 - Privacy as friction for state security, think China’s surveillance of Uighurs
 - But also: if we could listen to all conversation of decision makers, this would eliminate corruption, right?
 - In SciFi: “The Light of Other Days” by Arthur C. Clarke, Stephen Baxter, 2000

- Digital technology means we cannot see a panopticon building anymore
- The Panopticon is in our head, invisible. We are the guards.
- What are the consequences?
 - **Self-control**: adapt behaviors to follow the “norm”
 - **Self-censorship**: adapt opinions, speech to follow the “norm”
 - Think back to the early slides: why is privacy a human right? – Because it is the prerequisite for enjoying other human rights...
- See Xinjiang: “People were afraid to speak about their faith openly. Personal conversation about anything of substance seemed to cease.”²⁹

²⁹A. Chang and N. Fountain, “**Stuck In China’s Panopticon**,” *NPR*, Jul. 2019. [Online]. Available: <https://www.npr.org/2019/07/05/738949320/episode-924-stuck-in-chinas-panopticon>.

SUMMARY POINTS

1. We define privacy as access of one social actor to another, including what is revealed or concealed as well as what is sought or ignored. Privacy norms define expectations about appropriate access within a given context. Privacy management describes an actor's strategies and ability to regulate access to the self and others.
2. Privacy is related to social order through social control via the monitoring of actors, which can produce cooperation but also resistance and backlash.
3. Privacy has implications for cohesion because privacy norms dictate appropriate disclosure for different types of relationships and groups, and because privacy management shapes and is shaped by the intimacy of relationships.
4. Privacy is a resource that is unequally distributed in society, and the production and management of privacy may create inequality among social actors.
5. Much work on privacy focuses on first-order effects, that is, the impact of privacy on individuals. Sociological work is needed to examine the second-order effects, that is, the consequences of (changes in) privacy for the organization and functioning of society.

³⁰D. Anthony, C. Campos-Castillo, and C. Horne, “Toward a Sociology of Privacy,” *Annual Review of Sociology*, vol. 43, no. 1, pp. 249–269, 2017. DOI: 10.1146/annurev-soc-060116-053643.

PET LOVERS (WHY PRIVACY-ENHANCING TECHNOLOGIES?)

- Some commercial offerings depend on trust and privacy guarantees: e.g. personal DNA, adult sites
- But, unavoidably, very large-scale data breaches keep happening³¹ - the largest data leaks now affect **billions** of people!

The logo for CAM4, featuring the word "CAM" in black and "4" in orange.

March 2020: 10.88 billion records
Names, emails, chat transcripts, IP addresses

The logo for Yahoo!, featuring the word "yahoo!" in purple.

October 2017: 3 billion accounts
Account names, password hashes, security answers

The logo for AADHAAR, featuring a stylized fingerprint icon in red and yellow, with the word "AADHAAR" in red below it.

March 2018: 1.1 billion people
Names, ID numbers, bank details, biometric photos, thumbprints, retina scans

³¹UpGuard, *The 72 Biggest Data Breaches (Updated 2025)*, <https://www.upguard.com/blog/biggest-data-breaches>, 2025

³²K. Hill and H. Murphy, "Your DNA Profile is Private? A Florida Judge Just Said Otherwise," *The New York Times*, Nov. 2019. [Online]. Available: <https://www.nytimes.com/2019/11/05/business/dna-database-search-warrant.html>

PET LOVERS (WHY PRIVACY-ENHANCING TECHNOLOGIES?)

- Some commercial offerings depend on trust and privacy guarantees: e.g. personal DNA, adult sites
- But, unavoidably, very large-scale data breaches keep happening³¹ - the largest data leaks now affect **billions** of people!
- Personal data may also be processed in another country and be subject to **their** local laws³²

Your DNA Profile is Private? A Florida Judge Just Said Otherwise

Privacy experts say a warrant granted in Florida could set a precedent, opening up all consumer DNA sites to law enforcement agencies across the country.

"For police officers around the country, the genetic profiles that 20 million people have uploaded to consumer DNA sites represent a tantalizing resource that could be used to solve cases both new and cold. But for years, the vast majority of the data have been off limits to investigators. The two largest sites, Ancestry.com and 23andMe, have long pledged to keep their users' genetic information private [...] **Last week, however**, a Florida detective announced at a police convention that he had obtained a warrant to [...] search its full database of nearly one million users"

³¹UpGuard, *The 72 Biggest Data Breaches (Updated 2025)*, <https://www.upguard.com/blog/biggest-data-breaches>, 2025

³²K. Hill and H. Murphy, "Your DNA Profile is Private? A Florida Judge Just Said Otherwise," *The New York Times*, Nov. 2019. [Online]. Available: <https://www.nytimes.com/2019/11/05/business/dna-database-search-warrant.html>

PET LOVERS (WHY PRIVACY-ENHANCING TECHNOLOGIES?)

- Some commercial offerings depend on trust and privacy guarantees: e.g. personal DNA, adult sites
- But, unavoidably, very large-scale data breaches keep happening³¹ - the largest data leaks now affect **billions** of people!
- Personal data may also be processed in another country and be subject to **their** local laws³²
- **Permission-less privacy: How to self-defend, as individuals, or as processors?**

³¹UpGuard, *The 72 Biggest Data Breaches (Updated 2025)*, <https://www.upguard.com/blog/biggest-data-breaches>, 2025

³²K. Hill and H. Murphy, "Your DNA Profile is Private? A Florida Judge Just Said Otherwise," *The New York Times*, Nov. 2019. [Online]. Available: <https://www.nytimes.com/2019/11/05/business/dna-database-search-warrant.html>

WHERE PRIVACY-ENHANCING TECHNOLOGIES STARTED

- Early 1990s: US government classifies encryption software (RSA) as ammunition in order to restrict its export
 - early browsers could only be published with shortened key length
- The response of “Cypherpunk” activists³³:

“Privacy is necessary for an open society in the electronic age. [...] We cannot expect governments, corporations, or other large, faceless organizations to grant us privacy [...] We must defend our own privacy if we expect to have any. [...] Cypherpunks write code. We know that someone has to write software to defend privacy, and [...] we’re going to write it.”

(Cypherpunk Manifesto, 1993)

³³<https://activism.net/cypherpunk/manifesto.html>

WHY SUCH A “WHO OWNS PRIVACY” SECTION IN FDS?

- Main message: “Valuation of Privacy” is not self-evident nor granted, and may be questioned for good reasons (Prison Guards vs PET Lovers)
- Focusing on PET:
 - seems to take sides
 - may create moral dilemmas (“protect child pornography?”)
 - but may be critical for running society (“protect whistleblowers!”)
- No clear-cut answers. However, as Computer Scientists, we “own” part of the privacy discussion and owe society some answers:
 - ICT eroded (obliterated?) privacy for humanity
 - ICT could restore privacy, perhaps even has the obligation to do so ...
- We (CS) are not alone: studying + developing societal “control technologies” also in Law, Sociology